

IIT TIRUPATI · MOPR HACKATHON

GramDrain

LiDAR Pipeline Calculator

Ⓢ DROP YOUR FILE

Upload LAS / LAZ / ZIP (up to 10 GB)

Drag and drop file here

Limit 10GB per file • LAS, LAZ, ZIP

Browse files



Dhal_Hoshiarpur...

0.6GB

✓ AUTO-DETECTED FROM HEADER

Village DHAL_HOSHIARPUR_31235

Points 23.43 M

File size 679.5 MB

Coord type PROJECTED

X range 3546030.66 → 562803.17

Y range 301.47 → 562230.86

Z range 198.0 → 3546650.8 m

Ⓢ OVERRIDE (OPTIONAL)

> Edit parameters



Calculate Pipeline

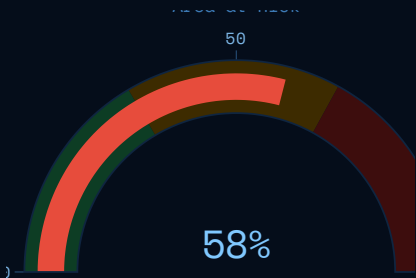
LiDAR Pipeline Calculator

GRAMDRAIN · IIT TIRUPATI · MOPR HACKATHON

✓ ✓ Pipeline complete!

DHAL_HOSHIAR
PUR_31235

EPSG:32643 · PROJECTED · 23.43M
PTS · 680 MB



↓
Download
JSON

1

INGEST

2

SCALE

3

GROUND

4

DTM

5

DERIVATIVES

6

ML

7

CLEANUP

8

ROUTING

9

EXPORT

STEP 01

Raw LiDAR Ingest

Point cloud ingestion & inspection

TOTAL
POINTS

23.43
M

raw:
23,431,282

FILE
SIZE

679.5
MB

0.66 GB

DENSITY

0.0
pt/m²

1676363.1985
km²

X EXTENT

29832
28 m

3546030.66 →
562803.17

Y EXTENT

56192
9 m

301.47 →
562230.86

COORD /
EPSG

PROJECTED

OK

EPSG:32643 ·
PROJECTED

STEP 02

Dynamic Scaling

Voxel grid + stride decimation

VOXEL SIZE

10.0 m

decimation cell

STRIDE

1 / 50

voxel keeper

EFFECTIVE
PTS

0.47M

post-decimation

GRID CELLS

56,194

×

298,324

16764.02M cells

GRID RAM

319748

MB

estimated peak

Point Decimation (millions)

Million Points

20
15
10
5
0

Raw Points

Effective Pts

STEP 03

Ground Classification

CSF cloth simulation filter

MODE

Regular
(CSF)

UNIQUE
CELLS

0.47M

filled voxel cells

GROUND PTS

0.30M

classified ground

GROUND %

65.0%

of decimated pts

OFF-GROUND
%

35.0%

veg/building

Ground / Off-Ground Split

Off-Ground
35%

Ground
65%

STEP 04

DTM Generation

Rasterise ground → COG

RESOLUTION

1.0

m/px

output GSD

RASTER DIMS

620 × 573

0.36M cells

Z MIN

198.0 m

above MSL

Z MAX

3546650

.8 m

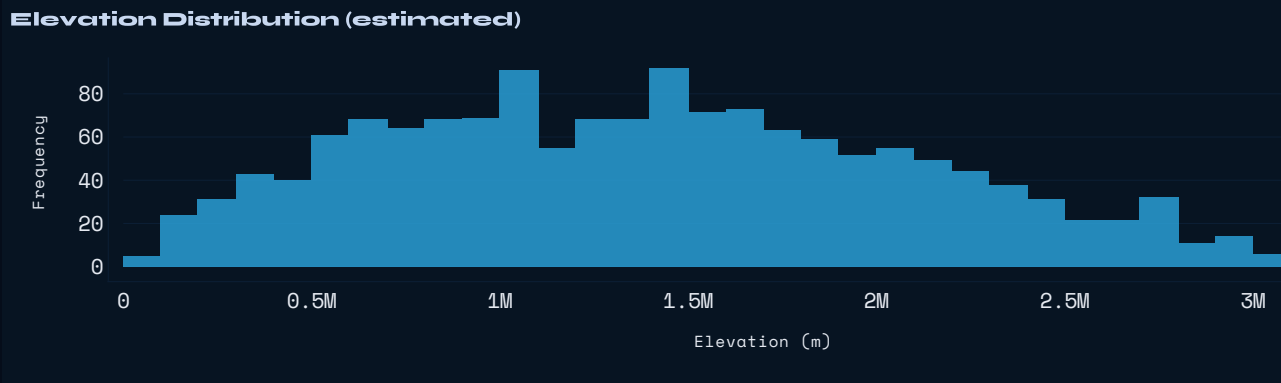
above MSL

Z RANGE

3546452

.8 m

Median 3×3 + Gaussian σ=1



STEP 05

Terrain Derivatives

Slope · Flow accumulation · TWI

SLOPE MIN

0.00°

flat terrain

SLOPE MAX

1.0°

steepest cell

STREAM THRESH

13,410,894,119

p97 accum value

STREAM CELLS

44,036

drainage network

TWI MIN

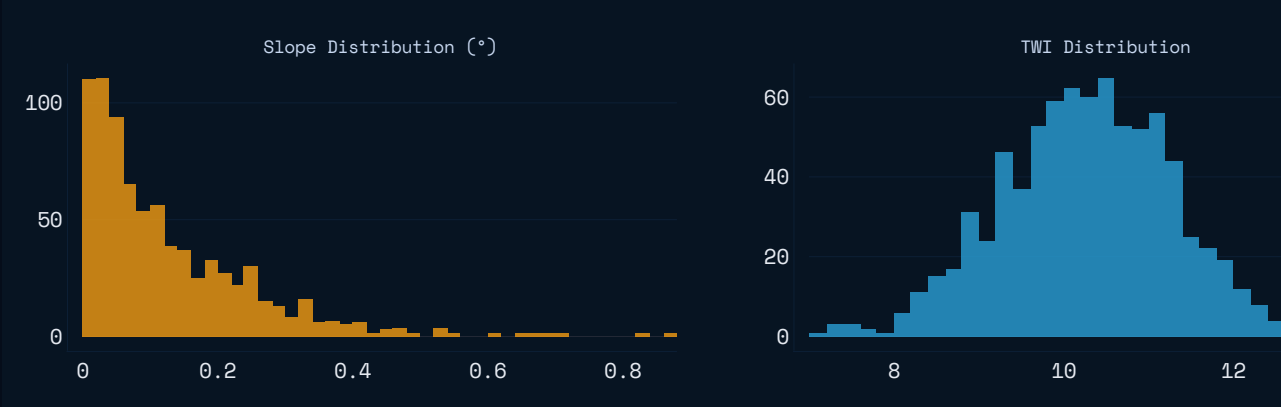
7.3

driest

TWI MAX

13.3

wettest



TWI Distribution

STEP 06

Risk Modelling

Rule-based labels → XGBoost 12-feature matrix

RULE HIGH

65,248

label=2train

RULE MEDIUM

148,057

label=1train

FEATURES

12

per cell

TRAIN SAMPLES

200,000

XGBoost fit

CONF MEAN

0.830

avg prob(high)

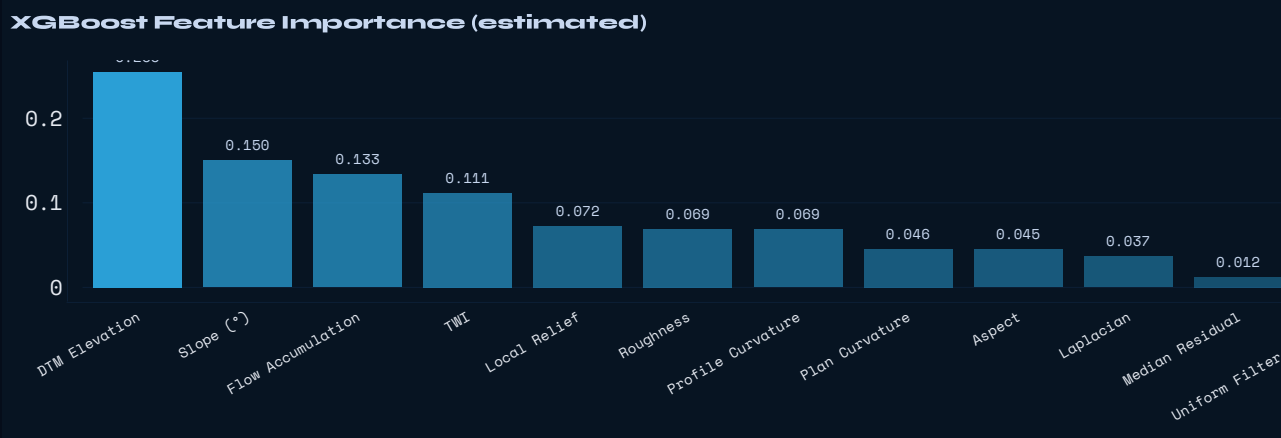
CONF MAX

0.910

peak confidence

Feature Matrix

Label Distribution



STEP 07

Morphological Post-processing

binary_opening 3x3 → binary_closing 5x5

NOISE
REMOVED

7,236

pixels erased

FINAL
HIGH

61,141

risk=2 cells

FINAL
MEDIUM

144,920

risk=1 cells

ML
FALLBACK

NO

ML used

NOISE
REDUCTION

3.4%

213,297 → 206,061



STEP 08

Drainage Network Design

Gravity-fed Dijkstra routing to natural streams

HOTSPOT
CLUSTERS

67

connected components

PROPOSED
DRAINS

4

gravity-routed paths

TOTAL
LENGTH

0.35 km

drain path distance

AVG
DRAIN

44.36 km

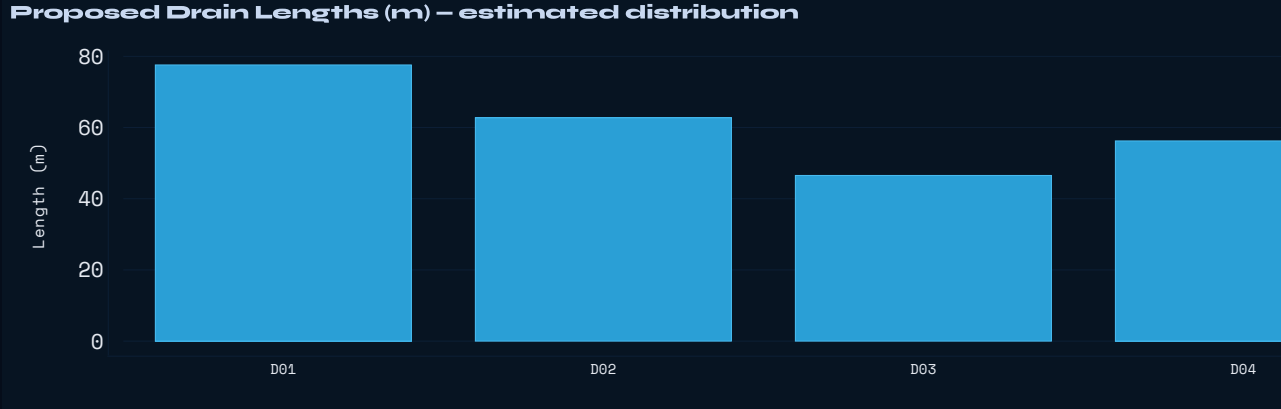
per drain

ROUTING ALGORITHM NOTE

Cost function:

(1-norm_accum)×60 + norm_elev×40 + 1

Low-cost corridors follow natural valleys (high accumulation + low elevation).
NODATA cells = 9999 (impenetrable) — drains never exit village boundary.
Only clusters ≥ 25 high-risk pixels are routed; max 50 clusters per village.



STEP 09

Export Outputs

OGC-compliant COG rasters · GeoPackage · Reports

☒ Pipeline finished — all output files are ready to download.

LAYER	FILENAME	FOLDER	STATUS
Ground LAS	DHAL_HOSHIARPUR_31235_ground.las	ground/	↓ Download
DTM COG	DHAL_HOSHIARPUR_31235_dtm_cog.tif	dtm/	↓ Download
Slope COG	DHAL_HOSHIARPUR_31235_slope_cog.tif	rasters/	↓ Download
Accumulation COG	DHAL_HOSHIARPUR_31235_accum_cog.tif	rasters/	↓ Download
TWI COG	DHAL_HOSHIARPUR_31235_twi_cog.tif	rasters/	↓ Download
Risk COG	DHAL_HOSHIARPUR_31235_risk_cog.tif	rasters/	↓ Download
Confidence COG	DHAL_HOSHIARPUR_31235_confidence_cog.tif	rasters/	↓ Download

GeoPackage

DHAL_HOSHIARPUR_31235_drainage.gpkg

gpkg/

Download

Village Report

DHAL_HOSHIARPUR_31235_report.png

village_reports/

Download

Full Parameter Sheet

Parameter	Value
CLEANUP	
Noise removed	7,236
Final high	61,141
Final medium	144,920
ML fallback	NO
ROUTING	
Hotspot clusters	67
Proposed drains	4
Drain length	0.35 km
Avg drain	44.36 km

GRAMDRAIN · IIT TIRUPATI MOPR HACKATHON · DHAL_HOSHIARPUR_31235 · 23,431,282 POINTS · 4 PROPOSED DRAINS · 0.35 KM